

Contact

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📍 Schwäbisch Hall, Germany

Skills

Programming

Python, R, SQL (intermediate)

C++, Linux (fundamental)

ML Frameworks

PyTorch, OpenCV, Scikit-learn,
Gymnasium, Scipy, Smolagents

Languages

German (native)

English (C1)

Certifications

Hugging Face:

Deep Reinforcement Learning

Fundamentals of LLMs

Fundamentals of Agents

Research Interests

Computer Vision

Reinforcement Learning

OpenClaw Tools

Agentic AI

Jannik Wirtheim

Data Science & Machine Learning

in LinkedIn  GitHub  Personal Website



Education

M.Sc. Social and Economic Data Science (1.2) 10/2023 – 03/2026
University of Konstanz

B.Sc. Business Psychology (1.3) 10/2020 – 09/2023
Stuttgart Technical University of Applied Sciences

Bank Clerk Apprenticeship (84/100 points) 09/2016 – 02/2019
VR Bank Heilbronn - Schwäbisch Hall e.G.

Experience

Research Assistant, Data Science and Generative AI since 06/2026
Fraunhofer IAO, Heilbronn

Contributing to projects at the intersection of Data Science and Generative AI.

Master's Thesis 10/2025 – 03/2026
Max Planck Institute of Animal Behavior, Konstanz

Imitation learning of multi-agent behavior in a competitive predator-prey environment from videos using generative adversarial imitation learning (GAIL).

- Developed a data preprocessing pipeline using YOLOv11 (object detection) and DeepSORT (multi-object tracking) to extract fish trajectories from videos.
- Learned a dense latent trajectory representation via a self-supervised autoencoder to capture meaningful behavioral patterns.
- Designed and trained a GAIL framework, optimized using evolutionary strategies in a vectorized environment.
- Conducted model evaluation and applied explainable AI methods to analyze emergent behavior and collective decision-making under survival pressure.

Student Assistant, Computer Science Department 09/2025 – 12/2025
Diak Klinikum Landkreis Schwäbisch Hall gGmbH

Contributed to computer science projects within a clinical environment, supporting both technical and administrative processes.

- Supported large-scale system migration (setup, data transfer, support).
- Assisted in IT infrastructure and maintenance across clinical departments.
- Assisted in the design and development of a web application.

Research Internship, Social and Economic Science 08/2022 – 09/2023
International Performance Research Institute gGmbH, Stuttgart

Conducted research in social and economic contexts with a strong focus on statistical analysis and data-driven methods.

- Developed a statistical reporting tool within the research project "FlyNet".
- Designed guidelines to improve data quality for predictive maintenance and machine learning pipelines within the project "MLready".

Student Assistant, Business Psychology Laboratory 09/2021 – 08/2022
Stuttgart Technical University of Applied Sciences

Supported research and operational tasks across scientific processes.

- Statistical support for qualitative and quantitative studies.
- Created video tutorials for biofeedback measurement using movisens.

Bank Clerk 02/2019 – 08/2019
VR Bank Heilbronn - Schwäbisch Hall e.G.

Gained experience in banking operations with focus on client relations and advisory.

- Handled front-office tasks (cash, scheduling, complaints).
- Advised customers on financial products and supported sales.